Online Experiments for Language Scientists

Lecture 6: Word learning

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Matter arising: office hour

- I am available for extra help (for if you are behind on labs only!) – **Thursday at 10am**

Matters arising from the self-paced reading lab

“Is there a way to write a long experiment that doesn’t involve copying, pasting and editing the same code over and over?”

- Yes, several! My preferred method: **functions that build trials**, coming in week 7, this week is the last of copy-and-paste code

“Is there a way to automatically randomise the order of things (e.g. buttons on a button-response trial)”

- Yes! Also coming in week 7

Reminder about Undergraduate Assessment 1

- Due on Thursday 12th November
- If you haven't already, read:
  - The assignment brief (https://kennysmithed.github.io/oels2026/assessment/UGAssignmentBrief2026.pdf)
  - The FAQ (https://kennysmithed.github.io/oels2026/assessment/oels_assignment_faq.html)
  - The generative AI policy (https://kennysmithed.github.io/oels2026/assessment/GenerativeAIPolicy.pdf)
- No questions after 11am on Monday 9th November!
  - People with extensions please extend this deadline appropriately
- For postgrad students only (i.e. MSc, PhD students): this assignment is optional and formative, i.e. doesn't contribute to your grade

Chua & Pan (2026)





Tabitha Chua *(National University of Singapore)*

Steven Pan *(National University of Singapore)*

Chua, T. J. E., & Pan, S. C. (2026). Duolingo-inspired pretesting with words and pictures improves vocabulary learning. *Cognitive Research, 11,* 20.

Four duolingo-like word learning experiments on Prolific

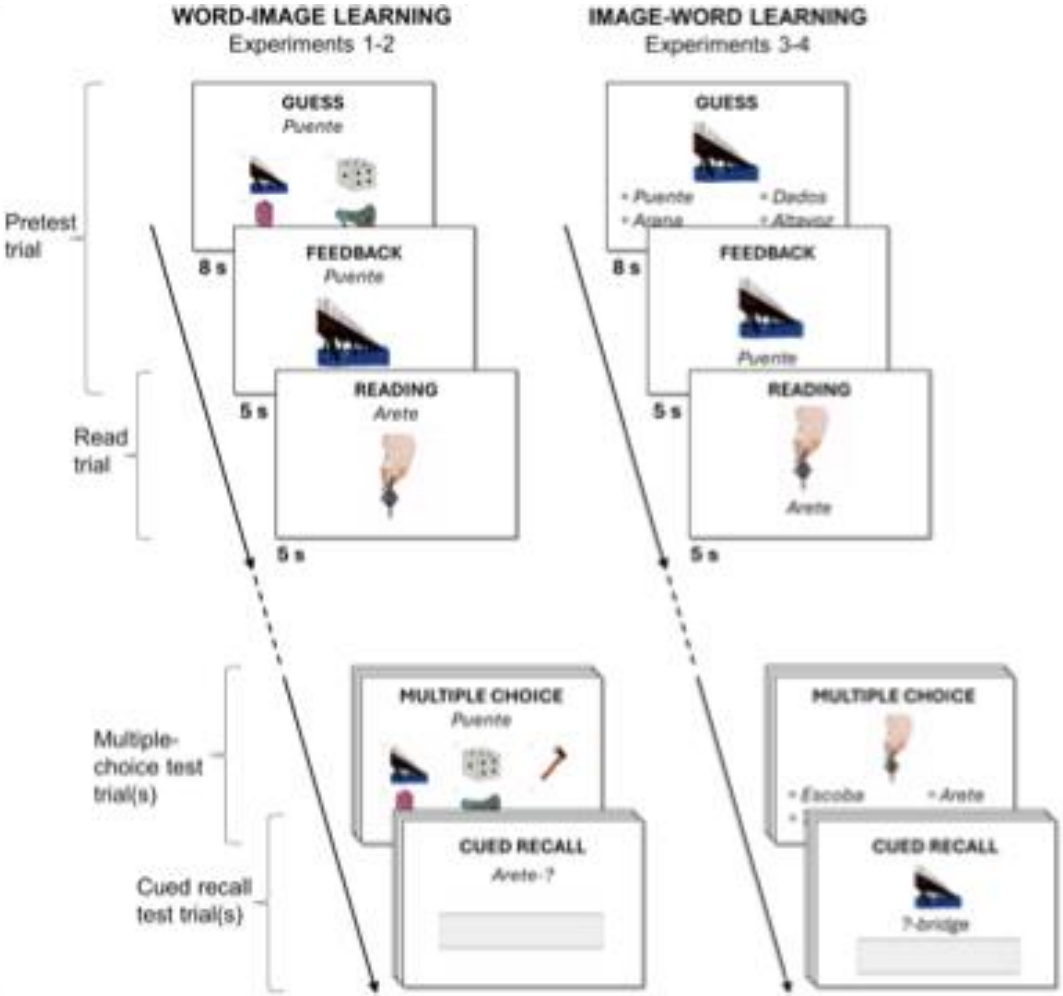
- Does **pretesting** (guess with feedback) facilitate word learning?

Procedure and design

Exp 1: word-to-image, training trial types randomly intermixed, test trial types randomly intermixed

Exp 2: word-to-image, **blocked vs randomized testing**

Exps 3-4: Same as 1-2, but image-to-word



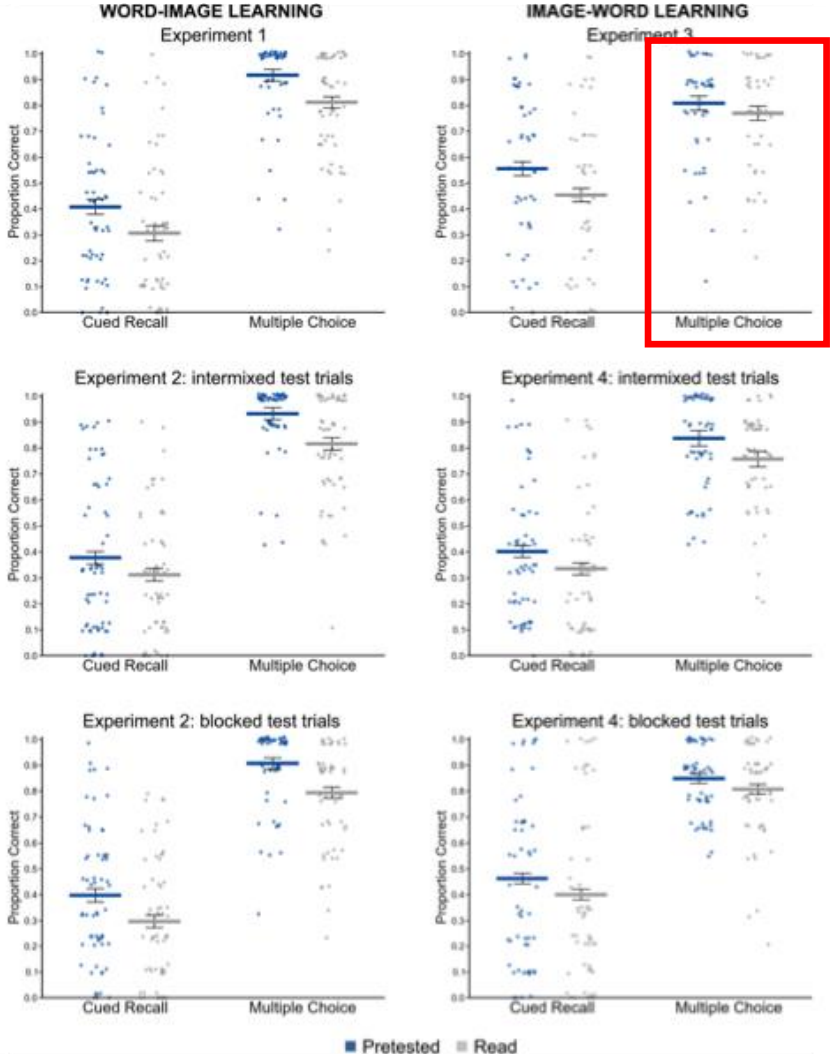
Word learning

Demo with this week's lab code

Image-to-word, blocked testing

Sample size, study duration etc

- Self-reported fluent speakers of English, resident in an English-speaking country, >95% Prolific approval, age 21-45
- N=78>58 (Exp 1), 153>123 (Exp 2), 65>47 (Exp 3), 162>113 (Exp 3)
- ?? minutes, $2.67


WORD-IMAGE LEARNING
IMAGE-WORD LEARNING
Experiment 1
Experiment 3
Experiment 2: intermixed test trials
Experiment 4: intermixed test trials
Experiment 2: blocked test trials
Experiment 4: blocked test trials
Proportion Correct
Cued Recall
Multiple Choice
Pretested
Read

Chua & Pan's conclusions

Pretesting does facilitate word learning

- For both word-to-image and image-to-word learning
- The effects seem not to depend on the details of test trial ordering

Additional note

- Successful use of online data collection for a task involving sustained attention and effort
- But there's always something to keep participants engaged

Additional time for Q&A/discussion on this week's reading

Coming up in labs: saving data to the server

Running experiments in the lab

Various ways to recruit participants to come into the lab (SONA, advertising on campus…)

Programming environment to create the experiment (PsychoPy, Eprime…)

Supporting files & data are saved on the lab's computer



Running experiments online

Participants are recruited to do experiments on their **own** devices

Programming environment to create the experiment (so it is presented on participants' devices)



HTTP posts, PHP saves

HTTP gets HTML, javascript, …

Supporting files & data are saved outside of participants' devices



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Data and computer security

Explainer

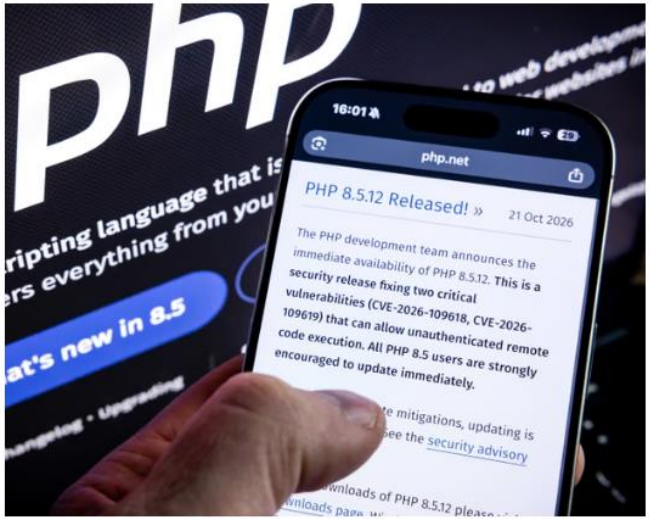
What is PHP and how serious is the computer language flaw causing website shutdowns?

An estimated 70% of websites use the open-source language. What is the identified vulnerability in it?

Josh Taylor *and* Robert Booth

Thu 22 Oct 2026 17.42 BST

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Vulnerabilities discovered in the programming language PHP could allow hackers to gain control over servers. Photograph: ANP/Shutterstock

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How it works

1. Our experiment code collates the data on the participant's computer
- jsPsych stores data for every trial

2. We send it to the server using fetch and POST

```
function simple_save_data(name, data_in){
    var url = 'save_data.php';
    var data_to_send = {filename: name, filedata: data_in};
    fetch(url, {
        method: 'POST',
        body: JSON.stringify(data_to_send),
        headers: new Headers({
                'Content-Type': 'application/json'
        })
    });
}
```

How it works

3. Our PHP script on the server processes and saves the incoming data

- This is the dangerous bit!
- save_data.php needs to include protections to prevent people running arbitrary commands on our server!
- The save_data.php we give you includes these (and there are substantially more than in previous versions!)

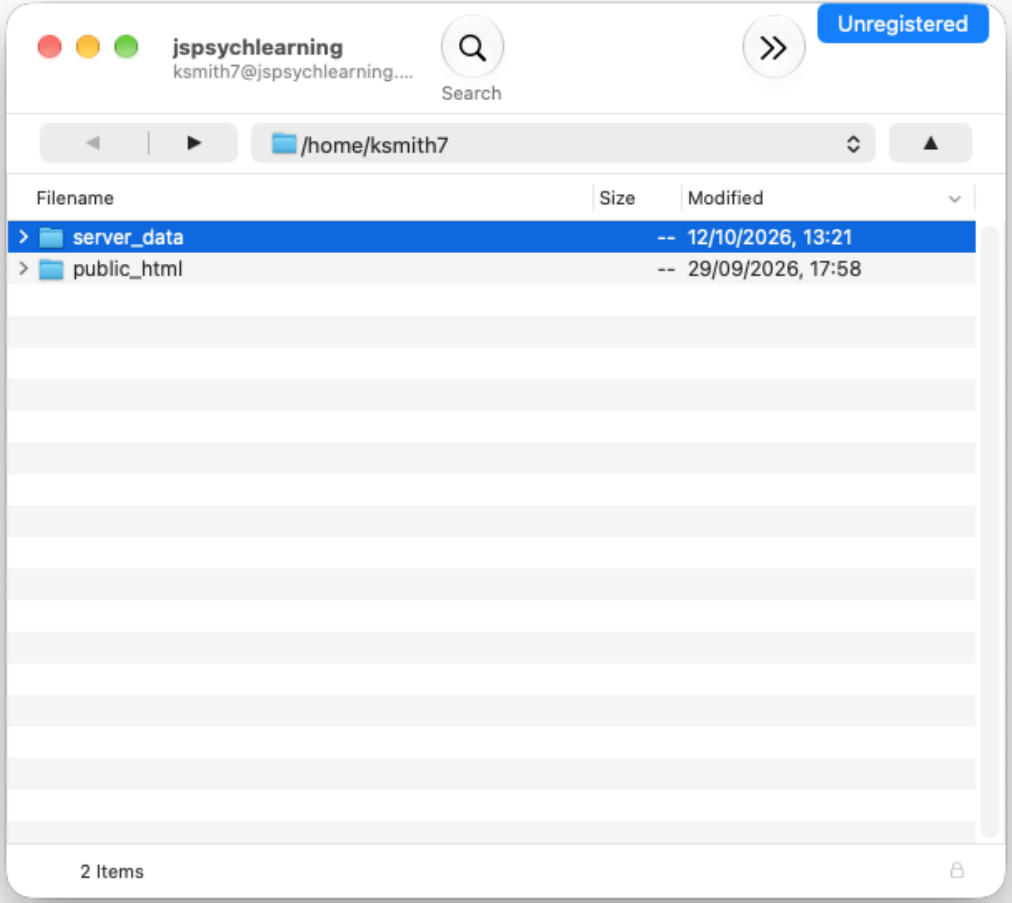
Where to save data

Not in public_html!

- Data usually shouldn’t be public

In server_data

- Our PHP script can write there
- But only you can access it



When to save data?

For now: at the end of the experiment, in a single file

```
var jsPsych = initJsPsych({
  on_finish: function () {
    var all_data = jsPsych.data.get(); //get all data
    var all_data_as_csv = all_data.csv(); //convert to csv format
    save_data("wordlearning_data.csv", all_data_as_csv); //save it using save_data function defined in utilities.js
  },
});
```

Better (week 8 on):

- **At the end of each trial**
- **Seperate file per participant**

Next up

Wednesday lab

- **Same (different) venue this week ! G.06, 50 George Square**

Next week:

- Frequency learning (reading is **technical**, see my notes re. not worrying too much about the later sections!)